



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.1

Division Expressions with Fractions and Whole Numbers

Lesson 6-1 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period

EVIDENCE LAB MISSION

The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact $\frac{1}{4}$ -foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.



TODAY'S OBJECTIVES

Content Objective: I can divide a whole number by a unit fraction and divide a fraction by a whole number by writing the whole number over 1 and using Keep, Change, Flip.

Language Objective: I can explain each step of Keep, Change, Flip using the words reciprocal, divisor, and quotient.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reciprocal

Quotient

Dividend

Divisor

Unit fraction

Multiplicative inverse

Tape diagram



Tap any word to see its meaning and a picture.

1

Flipping the numerator and denominator gives the

2

The answer to a division problem is the

3

In $6 \div 1/2$, the number 6 being divided is the

4

In $6 \div 1/2$, the number $1/2$ we divide by is the

5

A fraction with a numerator of 1, like $1/5$, is a

6

Two numbers whose product is 1 are

of each other.

7

A bar cut into equal parts to show sharing is a .

Write About the Math

How does interpreting fraction division help the detective?

¿Cómo ayuda interpretar la división de fracciones al detective?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Reciprocal**
Recíproco

☐ **Quotient**
Cociente

☐ **Dividend**
Dividendo

☐ **Divisor**
Divisor

☐ **Unit fraction**
Fracción unitaria

Model: *There will be more than 3 sections because each piece is smaller than a whole foot.* — Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can mark ____ sections because $6 \div 1/3 =$ ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Reciprocal
2. **Notes 2:** Quotient
3. **Notes 3:** Dividend
4. **Notes 4:** Divisor
5. **Notes 5:** Unit fraction
6. **Notes 6:** Multiplicative inverse
7. **Notes 7:** Tape diagram
8. **Watch for this mistake:** *A common mistake in Interpret Division of Fractions is assuming that dividing always makes the answer smaller. For $4 \div 1/3$, students reason 'dividing shrinks the number' and compute $4 \div 3 \approx 1.3$ instead of asking how many $1/3$ -foot pieces fit into 4 feet (each whole foot holds 3 thirds, so $4 \times 3 = 12$). Before you submit, check: if the divisor is a fraction smaller than 1, the quotient must be BIGGER than the dividend, not smaller.*
9. **Try It 1:** B. $2/3 \div 2$ — Splitting $2/3$ into 2 equal pieces is $2/3 \div 2$. Write the 2 as $2/1$, then Keep, Change, Flip: $2/3 \times 1/2 = 2/6 = 1/3$ yard.
10. **Try It 2:** — $3/4 \div 1/4$ asks how many one-fourth pieces make up three-fourths. The fraction bar shows $3/4$ already built out of fourth-size pieces, so counting those pieces gives the quotient: 3. Keep, Change, Flip agrees — $3/4 \times 4/1 = 12/4 = 3$. Neither number is a whole number here, so there is nothing to write over 1.